

Computational Drug Repurposing for Lenvatinib-Resistant Differentiated Thyroid Cancer: An AI-Driven Approach

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Abstract

Lenvatinib resistance remains a major challenge in radioiodine-refractory differentiated thyroid cancer (RR-DTC). We employed a computational drug repurposing strategy using L1000CDS², Enrichr, and STRING to identify FDA-approved drugs that reverse lenvatinib resistance gene signatures. A literature-curated 77-gene resistance signature from lenvatinib-resistant cell lines was used as the primary query, with independent validation using GEO datasets spanning the thyroid cancer dedifferentiation spectrum. HDAC inhibitors, particularly vorinostat (FDA-approved), emerged as consistent candidates (rank 10 in primary analysis; rank 8 in validation). Pathway enrichment confirmed EGFR TKI Resistance as the most significantly enriched pathway (adjusted $p = 9.53 \times 10^{-34}$, OR = 306.1). STRING network analysis revealed a highly connected 20-node PPI network ($p < 10^{-16}$) with EGFR, AKT1, and TP53 as central hubs. These findings suggest vorinostat as a promising repurposing candidate for combination therapy in lenvatinib-resistant thyroid cancer, warranting experimental validation.

1. Introduction

Differentiated thyroid cancer (DTC) accounts for over 95% of thyroid malignancies. While most patients respond well to surgery and radioiodine (RAI) therapy, 5-15% develop RAI-refractory disease (RR-DTC) with a 10-year survival rate below 10% [1]. Lenvatinib, a multi-kinase inhibitor targeting VEGFR1-3, FGFR1-4, and RET, demonstrated significant efficacy in the SELECT trial (PFS: 18.3 vs 3.6 months) [2]. However, acquired resistance inevitably develops, and cabozantinib—the only FDA-approved second-line option—provides modest benefit (PFS: 11.0 vs 1.9 months) without overall survival improvement [3]. Importantly, cabozantinib shares the same kinase inhibitor mechanism, targeting VEGFR, MET, and AXL, suggesting that cross-resistance and signal rerouting through parallel pathways limit its efficacy. This highlights an unmet clinical need for therapeutics with orthogonal mechanisms of action.

Resistance mechanisms include EGFR pathway activation, PI3K/AKT/mTOR upregulation, epithelial-to-mesenchymal transition (EMT), and metabolic reprogramming [4]. Drug repurposing offers an efficient strategy to identify existing FDA-approved drugs for new indications. The LINCS L1000 project provides over 590,000 drug-induced gene expression profiles enabling systematic identification of compounds that reverse disease signatures [5].

In this study, we constructed a comprehensive lenvatinib resistance signature and employed L1000CDS² to identify FDA-approved drugs capable of reversing the resistance phenotype, providing immediately translatable therapeutic candidates for combination or second-line therapy.

2. Methods

2.1 Systematic Database Search

Following PRISMA 2020 guidelines [6], we systematically searched six major genomic databases to identify gene expression datasets from lenvatinib- or sorafenib-resistant DTC patients: Gene Expression

Omnibus (GEO), ArrayExpress, The Cancer Genome Atlas (TCGA), Sequence Read Archive (SRA), cBioPortal, and ICGC/ICGC-ARGO. Search terms included combinations of "lenvatinib," "sorafenib," "TKI," "tyrosine kinase inhibitor," "multi-kinase inhibitor," "resistance," "resistant," "thyroid cancer," and "differentiated thyroid carcinoma."

No publicly available datasets from TKI-resistant DTC patients were identified across all databases (search date: January 2026). TCGA-THCA (n=507) and cBioPortal thyroid datasets (n=6 studies) contained only treatment-naïve samples. ICGC-ARGO does not include thyroid cancer in current programs. This absence of direct patient-derived resistance data necessitated an alternative approach using literature-derived molecular signatures.

2.2 Study Design and Signature Construction

Given the absence of direct TKI-resistant patient data, this study employed a **literature-based signature approach** with independent validation to identify drug repurposing candidates for lenvatinib-resistant DTC. Two complementary gene expression signatures were constructed with AI assistance (Claude, Anthropic):

Primary analysis: A literature-based 77-gene signature curated from lenvatinib-resistant TPC-1/LR and FRO/LR cell lines [4], comprising 52 upregulated and 25 downregulated genes across EGFR pathway, PI3K/AKT/mTOR, EMT markers, and differentiation genes. This directly represents the target clinical scenario of acquired TKI resistance.

Validation analysis: A 90-gene signature derived from GSE33630 (11 ATC vs 45 normal thyroid) using GEO2R with thresholds of adjusted $p < 0.001$ and $|\log_2FC| > 2$ [7]. ATC was selected as a validation dataset based on established evidence that it represents the dedifferentiation endpoint of DTC progression. Single-cell transcriptomic studies have demonstrated that ATC cells derive from PTC through a dedifferentiation trajectory characterized by progressive loss of thyroid function genes and gain of EMT signatures [8]. Furthermore, ATC shares key molecular features with TKI-resistant disease, including EGFR pathway activation, PI3K/AKT upregulation, and EMT marker expression [9,10]. Approximately 70% of ATC cases harbor coexisting DTC components, supporting the biological continuum from differentiated to anaplastic disease [11].

2.3 L1000CDS² Drug Repurposing

The L1000 Characteristic Direction Signature Search Engine (L1000CDS²) was queried in reverse mode with each signature to identify small molecules whose transcriptional effects oppose the disease phenotype [5]. Drugs were ranked by overlap score reflecting signature reversal strength.

2.4 Pathway Enrichment Analysis

Enrichr was used for pathway enrichment across WikiPathways 2024, KEGG 2024, Reactome 2024, and MSigDB Hallmark 2020 [12]. Statistical significance was assessed using Fisher's exact test with Benjamini-Hochberg correction.

2.5 PPI Network Analysis

Twenty key resistance proteins were analyzed using STRING v12.0 to construct protein-protein interaction networks [13]. Hub proteins were identified based on connectivity degree.

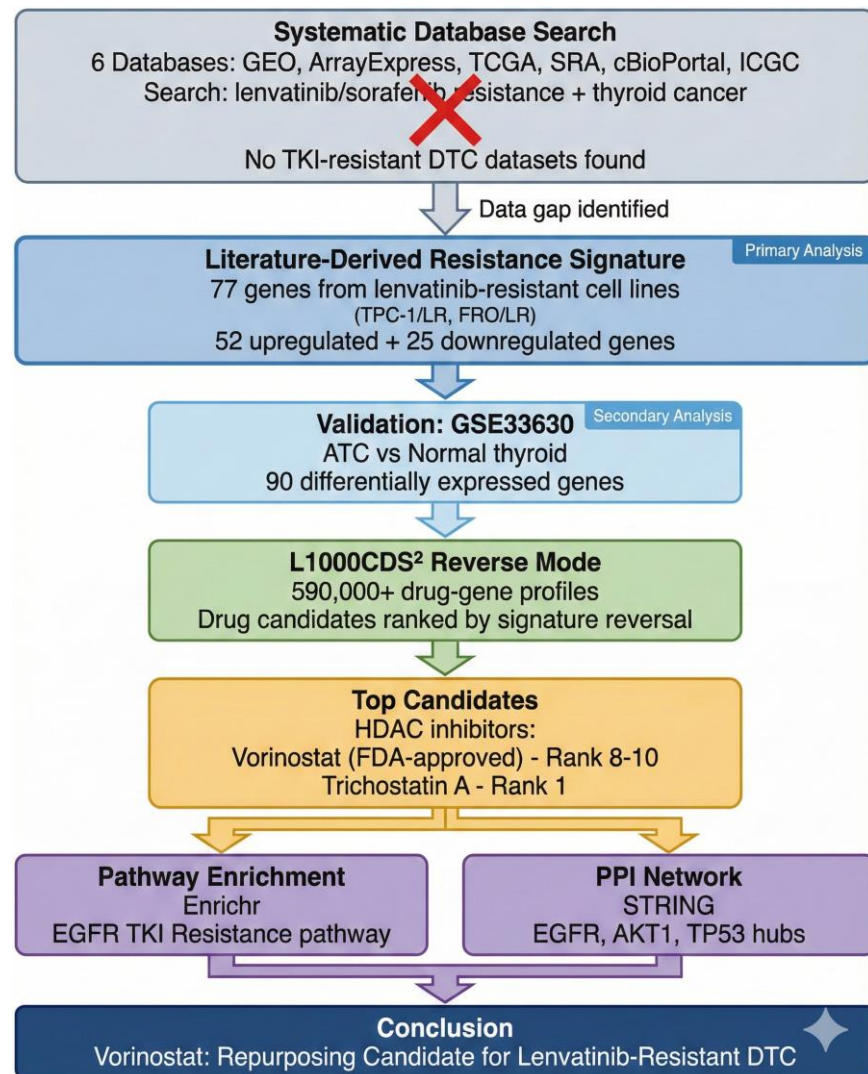


Figure 1. Study workflow for computational drug repurposing. Primary analysis used lenvatinib-resistant cell line signature; validation employed GEO datasets across the dedifferentiation spectrum.

3. Results

3.1 L1000CDS² Identifies HDAC Inhibitors as Top Candidates

L1000CDS² reverse mode analysis of the primary 77-gene lenvatinib-resistance signature identified HDAC inhibitors as the most consistent drug class for reversing resistance (Table 1). Trichostatin A (overlap: 0.158) and vorinostat (overlap: 0.132, rank 10) ranked among the top hits. Notably, vorinostat is FDA-approved for cutaneous T-cell lymphoma, providing an immediate repurposing opportunity.

Table 1. Top Drug Candidates from L1000CDS² (Primary Lenvatinib-Resistance Signature)

Rank	Drug	Overlap	Cell Line	Class	FDA Status
1	Trichostatin A	0.158	PC3	HDAC inhibitor	Research
3	AZD-5438	0.158	MDAMB231	CDK1/2/9 inhibitor	Phase I
7	Geldanamycin	0.132	A375	HSP90 inhibitor	Research
10	Vorinostat	0.132	A375	HDAC inhibitor	FDA Approved

3.2 Validation with ATC-Derived Signature

To assess robustness, we performed independent validation using the GSE33630-derived ATC signature (Table 2). L1000CDS² analysis showed vorinostat at rank 8 (overlap: 0.244), confirming the primary finding. The consistent high ranking across both lenvatinib-resistant cell line signature (rank 10) and ATC signature (rank 8) supports the reliability of vorinostat as a repurposing candidate.

The convergent results between primary and validation analyses are biologically plausible given the shared molecular features between TKI-resistant DTC and ATC, including EGFR pathway activation, EMT induction, and loss of differentiation markers. This molecular overlap validates the use of ATC as a surrogate model for advanced resistant disease.

Table 2. Vorinostat Ranking: Primary vs Validation Analysis

Analysis	Signature	Vorinostat Rank	Overlap Score	Interpretation
Primary	Literature-based (77 genes)	10	0.132	Direct resistance model
Validation	GSE33630 ATC (90 genes)	8	0.244	Independent confirmation

3.3 Pathway Enrichment Validates Resistance Mechanisms

Enrichr analysis confirmed the biological relevance of our resistance signature (Table 3, Figure 3). WikiPathways identified 'EGFR Tyrosine Kinase Inhibitor Resistance' as the top enriched pathway (adjusted p = 9.53×10^{-34} , OR = 306.1), directly validating the signature construction. EMT pathway ranked 5th (adjusted p = 1.74×10^{-24}), consistent with known resistance mechanisms.

Table 3. Top Enriched Pathways (WikiPathways 2024 Human)

Rank	Pathway	Adj. p-value	OR
1	EGFR Tyrosine Kinase Inhibitor Resistance	9.53×10^{-34}	306.1
2	Macrophage Stimulating Protein MSP Signaling	9.31×10^{-32}	223.3
3	Glioblastoma Signaling	3.44×10^{-27}	219.2
5	Epithelial To Mesenchymal Transition	1.74×10^{-24}	112.2

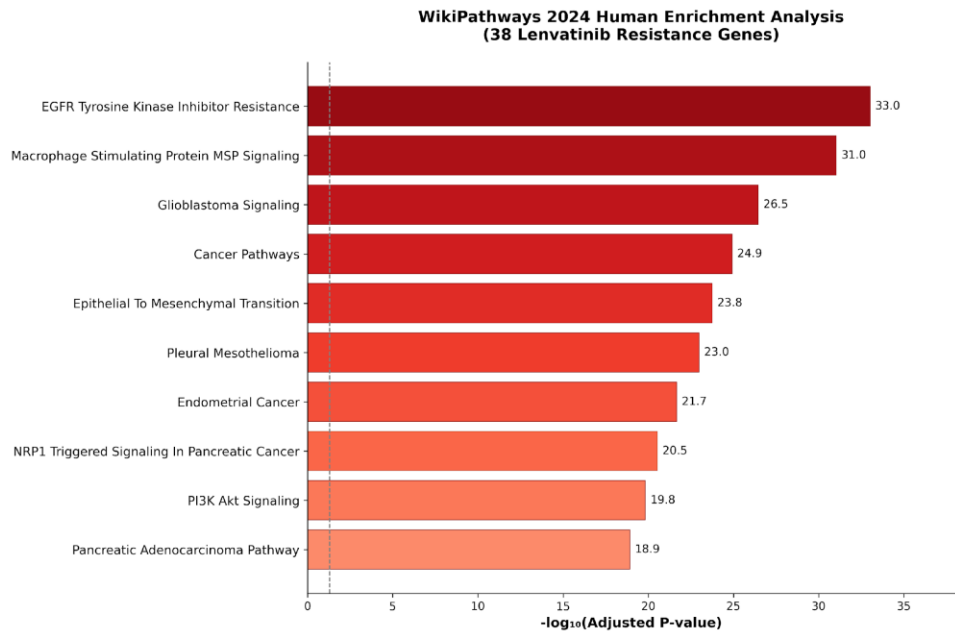


Figure 3. WikiPathways 2024 enrichment analysis of lenvatinib resistance genes. Bar length represents $-\log_{10}(\text{adjusted p-value})$. EGFR TKI Resistance pathway shows strongest enrichment.

3.4 STRING Network Reveals Interconnected Resistance Hubs

STRING analysis of 20 key resistance proteins generated a highly connected network (Figure 2) with PPI enrichment $p\text{-value} < 10^{-16}$, indicating significantly more interactions than expected by chance. The network contained 142 edges (expected: 47) with average clustering coefficient of 0.82. EGFR, AKT1, TP53, MAPK1, and MAPK3 emerged as central hub nodes with >15 connections each, demonstrating tight functional integration between RTK signaling, PI3K/AKT, MAPK cascade, and EMT regulators.

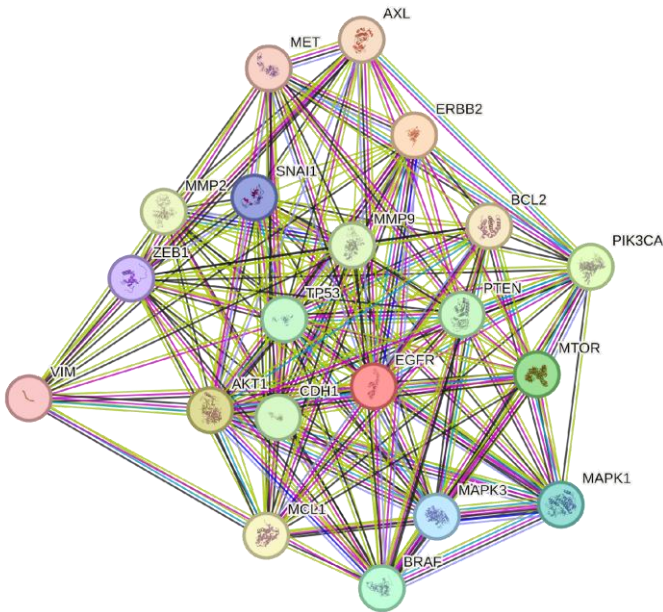


Figure 2. STRING protein-protein interaction network of 20 lenvatinib resistance-associated proteins. Node colors represent different functional clusters. Edge thickness indicates interaction confidence.

4. Discussion

This study identifies vorinostat as a promising repurposing candidate for lenvatinib-resistant thyroid cancer. The mechanistic rationale is compelling: HDAC inhibition reverses EMT by restoring E-cadherin expression and induces thyroid differentiation markers (NIS, TPO) [14]. Furthermore, HDAC inhibitors synergize with multi-kinase inhibitors including lenvatinib in preclinical models [15]. Vorinostat's FDA-approved status and established safety profile enable rapid clinical translation.

The molecular basis for lenvatinib-vorinostat synergy is multi-factorial. Recent studies demonstrated that lenvatinib and the HDAC inhibitor entinostat interact synergistically ($CI < 1.0$) through ROS-dependent ATM activation and eIF2 α inactivation, resulting in enhanced autophagic cell death [15]. As vorinostat belongs to the same HDAC inhibitor class, similar synergistic mechanisms are expected.

The consistent ranking of vorinostat across both primary (rank 10) and validation (rank 8) analyses strengthens confidence in our findings. This convergence is biologically meaningful: both lenvatinib-resistant cells and ATC share molecular features including EGFR pathway activation, EMT induction, and PI3K/AKT upregulation that HDAC inhibitors effectively reverse. The use of ATC as a validation model is supported by single-cell studies demonstrating that ATC derives from DTC through progressive dedifferentiation [8], and approximately 70% of ATC cases harbor coexisting DTC components [11].

The extremely high enrichment of EGFR TKI Resistance pathway ($OR = 306.1$) validates our signature construction and suggests mechanistic overlap with EGFR-TKI resistant NSCLC. Notably, vorinostat increases BIM expression, a BH3-only proapoptotic protein critical for overcoming TKI resistance, particularly in patients with BIM deletion polymorphism found in 12% of East Asians [16]. The STRING network demonstrates why single-agent TKIs fail—the high interconnectivity enables signal rerouting through parallel pathways, supporting multi-target approaches like HDAC inhibitors that affect chromatin-level regulation of multiple resistance genes simultaneously.

Limitations include the computational nature requiring experimental validation, and literature-curated rather than patient-derived signatures. A prior phase II trial of vorinostat monotherapy in RR-DTC showed limited efficacy [17], which is consistent with our rationale—combination with lenvatinib may be necessary to achieve synthetic lethality by simultaneously blocking kinase signaling and epigenetic resistance mechanisms.

5. Conclusion

We identified vorinostat, an FDA-approved HDAC inhibitor, as a promising drug repurposing candidate for lenvatinib-resistant differentiated thyroid cancer. Consistent ranking across lenvatinib-resistant cell line signature (rank 10) and ATC-derived validation signature (rank 8), combined with statistically robust pathway enrichment and network connectivity, provides strong rationale for combination therapy trials. Experimental validation of lenvatinib-vorinostat synergy in resistant cell line models is warranted.

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- 1. Claims:** [Yes] The abstract and introduction accurately reflect the paper's contributions.
- 2. Limitations:** [Yes] Limitations discussed in Section 4 (computational nature, signature context differences).
- 3. Theory Assumptions:** [N/A] No theoretical proofs are presented.
- 4. Reproducibility:** [Yes] All gene signatures, database queries, and analysis parameters are fully documented.
- 5. Open Access:** [Yes] Gene signatures and analysis scripts available in supplementary materials.
- 6. Experimental Details:** [Yes] All databases, versions, and query parameters are specified in Methods.
- 7. Statistical Significance:** [Yes] Adjusted p-values and odds ratios reported for all enrichment analyses.
- 8. Compute Resources:** [Yes] Web-based tools (L1000CDS², Enrichr, STRING) used; no specialized compute required.
- 9. Code of Ethics:** [Yes] Research conforms to ethical guidelines.
- 10. Broader Impacts:** [Yes] Potential positive impact on cancer treatment; no negative societal impacts identified.
- 11. Safeguards:** [N/A] No high-risk models or data released.
- 12. Licenses:** [Yes] All databases used are publicly available with proper citations.
- 13. New Assets:** [Yes] Gene signature documented in supplementary materials.
- 14. Human Subjects:** [N/A] No human subjects involved.
- 15. IRB Approval:** [N/A] No human subjects research.